Foundations of Time Series Analysis

Chapter 1

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Outline

- What is time-series data?
- The White Noise
- Difference equations
- The ACF and PACF
- Stationarity
- Testing for and fixing stationarity

What is time-series?

Section 1

What is time-series data?

- Data is connected across time — data has “memory”
- The order of the observation matters; the order has information in itself
  - $(y_{t-1}, y_t) \neq (y_t, y_{t-1})$
- In cross-sectional work, independence is ideal, and the order of the observations is unimportant
  - $(y_1, y_2) = (y_2, y_1)$

What is time-series data?

- What's forecasting
  - What forecasting is **not**
    - Prediction: Cross-sectional data; there is no time to project data through
    - Causal inference: A forecast does not require a causal relationship — you can advise about your future to a fortune teller, who will give you a forecast
  - What forecasting **is**
    - The conditional expectation $\mathbb{E}[y_{T+h}|\mathcal{F}_T]$
    - The expected value of y, h periods ahead, conditional on all the information and knowledge today ($\mathcal{F}_T$)

What is time-series data?

- Three types of forecast
  - Point forecast
    - A single number; our best guess of y_{T+h}
    - $\hat{y}_{T+hT} = \mathbb{E}[y_{T+h}|\mathcal{F}_T]$
  - Interval forecast
    - A range $[L_{T+h}, U_{T+h}]$ containing the true value with some stated probability (95%, ...)
    - $\hat{y}_{T+h|T} \pm z_{\alpha/2} \cdot \hat{\sigma}_{T+h|T}$
  - Density forecast
    - The full probability distribution in the interval forecast: $\hat{p}(y_{T+h}|\mathcal{F}_T)$

Forecasts are **not** about point predictions. Forecasts are about a “likely” range of values. The main role of the point estimate is to calibrate the location of the interval

What is time-series data?

- Why is the economic forecast so difficult?
  - **Structural changes**: Economic relationships aren't physical laws; they shift as institutions, technology, and policy change
  - **Measurement error and revisions**: Today's GDP release is not the number we'll have in two years
  - **The Lucas critique**: Agents *react* to policy; reduced-form relationships shift when the policy regime shifts
  - **Deep parameters**: The only true stable objects are primitives (preferences, technology) — not the estimated coefficients that depend on them

What is time-series data?

- The **loss function**
  - There are ∞ ways to forecast y_t. How are they evaluated? What's the benchmark?
  - The **loss function** is what evaluates the forecast
    - Let $e_{T+h} = y_{T+h} - \hat{y}_{T+h|T}$ be the "loss" or forecast error
  - Then:
    - **Squared loss:** $\mathcal{L}(e) = e^2$ — symmetric, penalizes large errors disproportionally
    - **Absolute loss:** $\mathcal{L}(e) = |e|$ — symmetric, penalty line in error size
    - **Asymmetric loss:** Over- and under-forecasting cost differently; i.e., deflation is considered more costly than inflation

What is time-series data?

- Squared loss: The Mean
  - Minimize $e^2 = \mathbb{E}[(y - \hat{y})^2]$
    - $\frac{d}{d\hat{y}}\mathbb{E}[(y - \hat{y})^2] = \mathbb{E}[y^2 - 2\hat{y}y + \hat{y}^2] = -2\mathbb{E}[y] + 2\hat{y}$
    - Make it equal to zero and solve for $\hat{y}$
    - $-2\mathbb{E}[y] + 2\hat{y} = 0$
    - $\hat{y} = \mathbb{E}[y]$

What is time-series data?

- Absolute loss: The Median
  - Minimize $|e| = \mathbb{E}[|y - \hat{y}|] = \int_{-\infty}^{\hat{y}} (\hat{y} - y) f(y) dy + \int_{\hat{y}}^{\infty} (y - \hat{y}) f(y) dy$
  - $\frac{d}{d\hat{y}} \mathbb{E}[|y - \hat{y}|] = F(\hat{y}) - [1 - F(\hat{y})] = 2F(\hat{y}) - 1$
    - Make it equal to zero and solve for $\hat{y}$
    - $2F(\hat{y}) - 1 = 0$
    - $F(\hat{y}) = \frac{1}{2}$

What is time-series data?

Squared loss penalizes errors quadratically — the optimum gets pulled toward the mean to avoid large errors.
Absolute loss penalizes errors proportionally — the median is robust to how far outliers sit from the center

Specifying the loss function is **not** a technical afterthought. It's a modeling choice tied to the actual decision problem.

The White Noise

Section 2

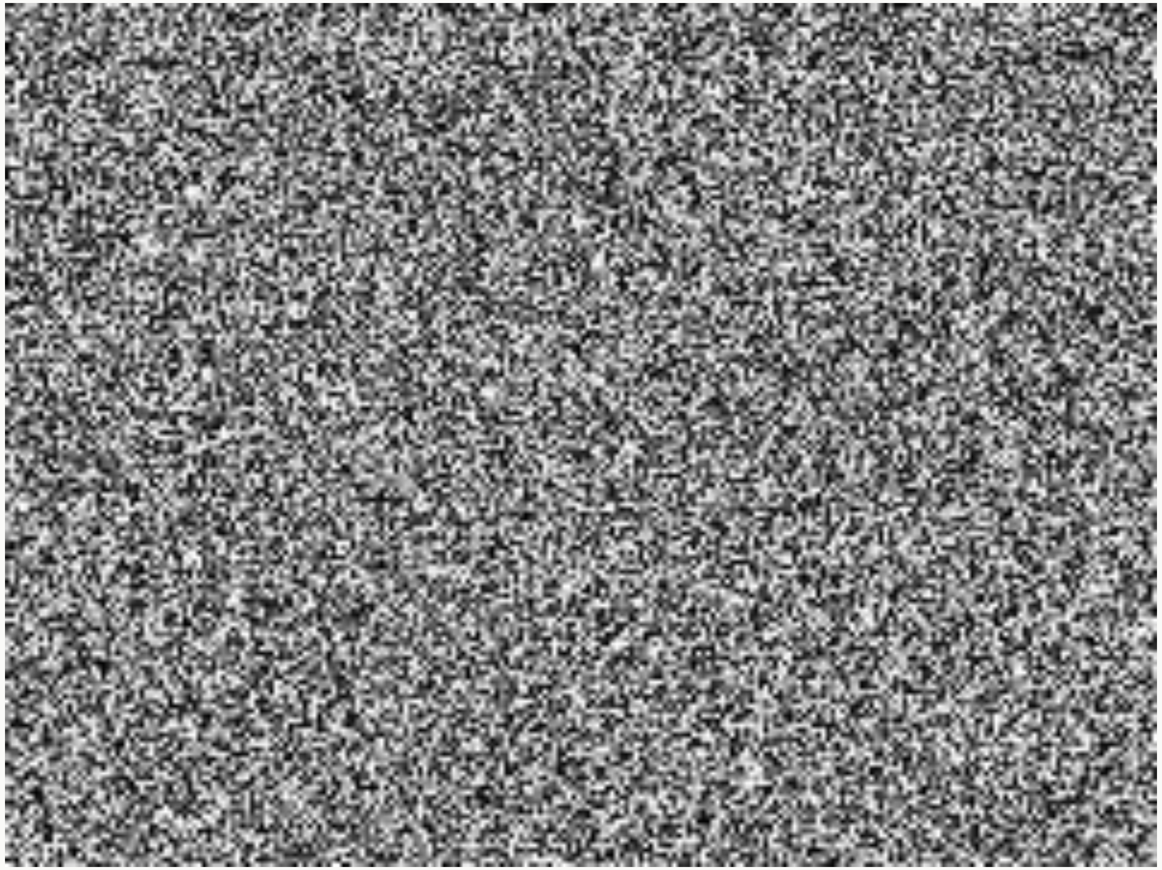
The White Noise

- Some time-series notation
  - The time index t runs over $\mathcal{T} = \{1, \ldots, T\}$; this is your sample
  - Arithmetic on t is meaningful
    - $y_t - y_{t-1}$ is the change
    - y_{t-12} is a year ago in monthly data
    - y_{T+h} is h periods beyond the sample
  - Statistical notation
    - $\mu = \mathbb{E}[y_t]$
    - $\gamma(h) = Cov(y_t, y_{t-h})$
    - $\rho(h) = Corr(y_t, y_{t-h})$
    - $\sigma^2 = Var(y_t)$

The White Noise

- Some time-series notation (cont…)
  - The **lag** operator
    - $Ly_t = y_{t-1}$
    - $L^k y_t = y_{t-k}$
  - The **difference** operator
    - $\Delta y_t = y_t - y_{t-1} = (1 - L)y_t$

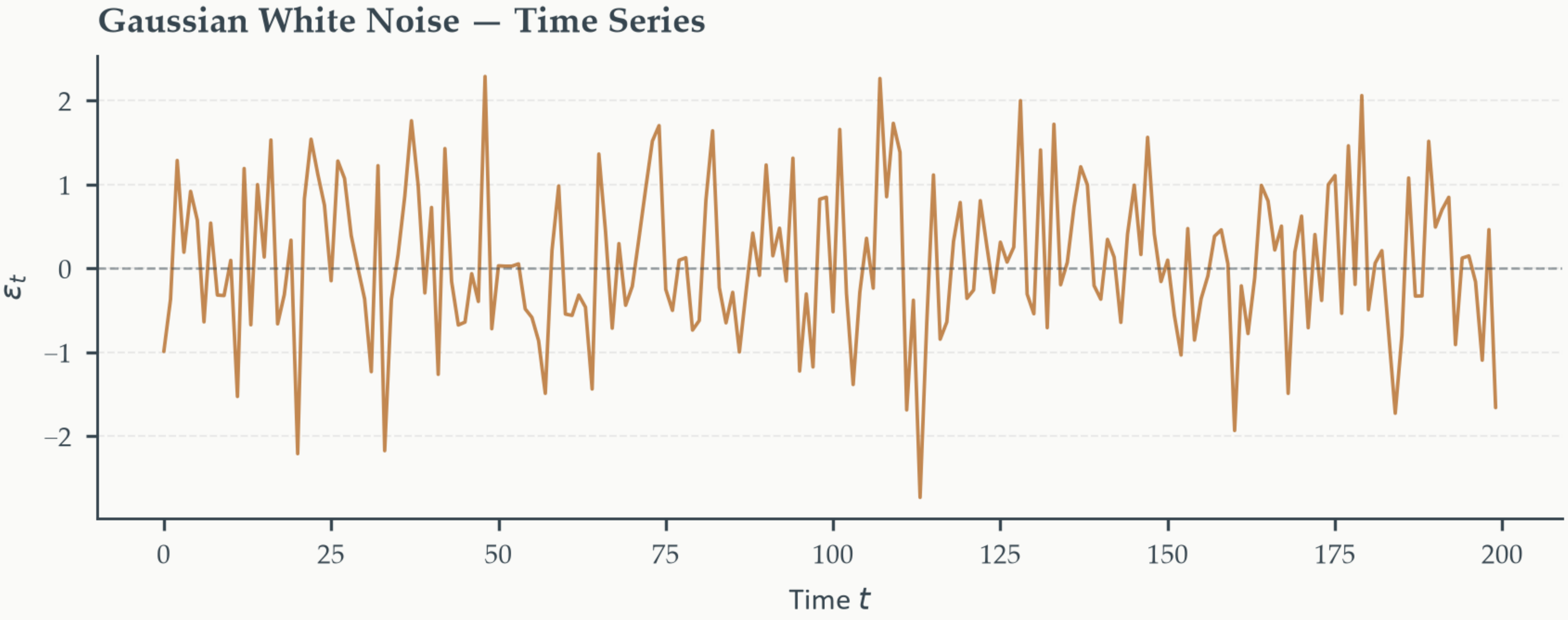
The White Noise



The White Noise

- The simplest time series — a baseline model
  - A $WN(0, \sigma^2)$ is a process with
    - Zero mean
    - Constant variance
    - Zero autocovariance at every nonzero lag
  - What behaves as WN?
    - An error term
    - Residuals of a properly run regression
    - If a model's residuals don't look like a WN, there is still information to extract and add to the model.

The White Noise



The White Noise

- WN and independence
  - Independence rules out any statistical relationship of any kind, not just linear ones
  - IID: Independent and Identically Distributed
    - Every data point is **independent** of the others, but they come from the same probability distribution and, therefore, share statistical properties

$$IID \text{ noise} \Rightarrow \text{Strong } WN \rightarrow \text{Weak } WN$$

Difference equations

Section 3

Difference equations

- If data has memory, we expect y_t to be a function of its own past
  - $y_t = f\left(y_{t-j}\right); j \geq 1$
- Add a memory parameter, a constant, and a WN error term
  - $y_t = \phi y_{t-1} + c + \varepsilon_t$
  - In loose terms, think of
    - c as the constant (pull) value y_t is equal to if there is no memory at all
    - ϕ as the memory (push) factor that keeps y_t closer to y_{t-1} instead of c

Difference equations

- Go back on time until initial **given** value y_0
  - $y_1 = \phi y_0 + c$
  - $y_2 = \phi y_1 + c = \phi^2 + (1 + \phi)c$
  - $y_3 = \phi y_2 + c = \phi^3 + (1 + \phi^2 + \phi)c$
  - Generalize to t
  - $y_t = \phi y_{t-1} + c = \phi^t y_0 + (1 + \phi^{t-1} + \cdots + \phi)c$

The **value** of ϕ governs the behavior and stability of $\{y_t\}$

Difference equations

- Series convergence
  - AR(1): 1 lag
    - $y_t = \phi y_{t-1} + c$
    - In "equilibrium", $y_t = y_{t-j} \forall j \neq 0$
    - Then
      - $y_t = \phi y_t + c$
      - $\bar{y} = \frac{c}{1-\phi}$
    - In terms of y_0
      - $y_t = \left(y_0 - \frac{c}{1-\phi}\right)\phi^t + \frac{c}{1-\phi}$
  - AR(2): 2 lags
    - $y_t = \phi_1 y_{t-1} + \phi_2 y_{t-2} + c$
    - In "equilibrium", $y_t = y_{t-j} \forall j \neq 0$
    - Then
      - $y_t = \phi_1 y_t + \phi_2 y_{t-2} + c$
      - $\bar{y} = \frac{c}{1-\phi_1-\phi_2}$
    - In terms of y_0
      - $y_t = \left(y_0 - \frac{c}{1-\phi_1-\phi_2}\right)\phi^t + \frac{c}{1-\phi_1-\phi_2}$

y_t is a weighted sum of the **initial deviation** from the long-run level of $\bar{y}$, decaying at rate ϕ^t, and the long-run level itself

Difference equations

- Model stability — AR(1)
  - $|\phi| < 1$ → **stable**: $y_t \to \bar{y}$
  - $|\phi| = 1$ → **unit root**: shocks have a permanent effect, there is no returning to $\bar{y}$
  - $|\phi| > 1$ → **explosive**: deviations growth without bounds

  - $\bar{y}$ exists only if $\{y_t\}$ is stable, that is, if $|\phi| < 1$

Difference equations

- Obtaining the **roots**
  - What is the function that governs the behavior of $\{y_t\}$?
    - We don't know. If we did, we wouldn't need to do any of this.
  - Assume then that $y_t = A\lambda^t$
    - An exponential function captures the behavior of many economic variables (compounding effects) and is very adaptable by changing A and λ

Difference equations

- Obtaining the **roots** (cont…)
  - AR(1)
    - $y_t = \phi y_{t-1} + c = y_t^h + y_t^p = \text{homogeneous} + \text{particular}$
    - Solve the **homogeneous** equation
      - $y_t - \phi y_{t-1} = 0$
      - $A\lambda^t - \phi A\lambda^{t-1} = 0$
      - $\lambda = \phi$
    - Solve the **particular** equation
      - Try $y_t^p = k$
      - $k = \phi k + c$
      - $k = \frac{c}{1-\phi}; \phi \neq 1$

Difference equations

- Obtaining the **roots** (cont...)
  - AR(2)
    - $y_t = \phi_1 y_{t-1} + \phi_2 y_{t-2} + c$
    - Solve the **homogeneous** equation
      - $y_t - \phi_1 y_{t-1} - \phi_2 y_{t-2} = 0$
      - $A\lambda^t - \phi_1 A\lambda^{t-1} - \phi_2 A\lambda^{t-2} = 0$
      - $\lambda^2 - \phi_1\lambda - \phi_2 = 0$
      - $\lambda = \frac{\phi_1 \pm \sqrt{\phi_1^2 - 4\phi_2}}{2} = (\lambda_1, \lambda_2)$
    - Solve the **particular** equation
      - Try yourself...

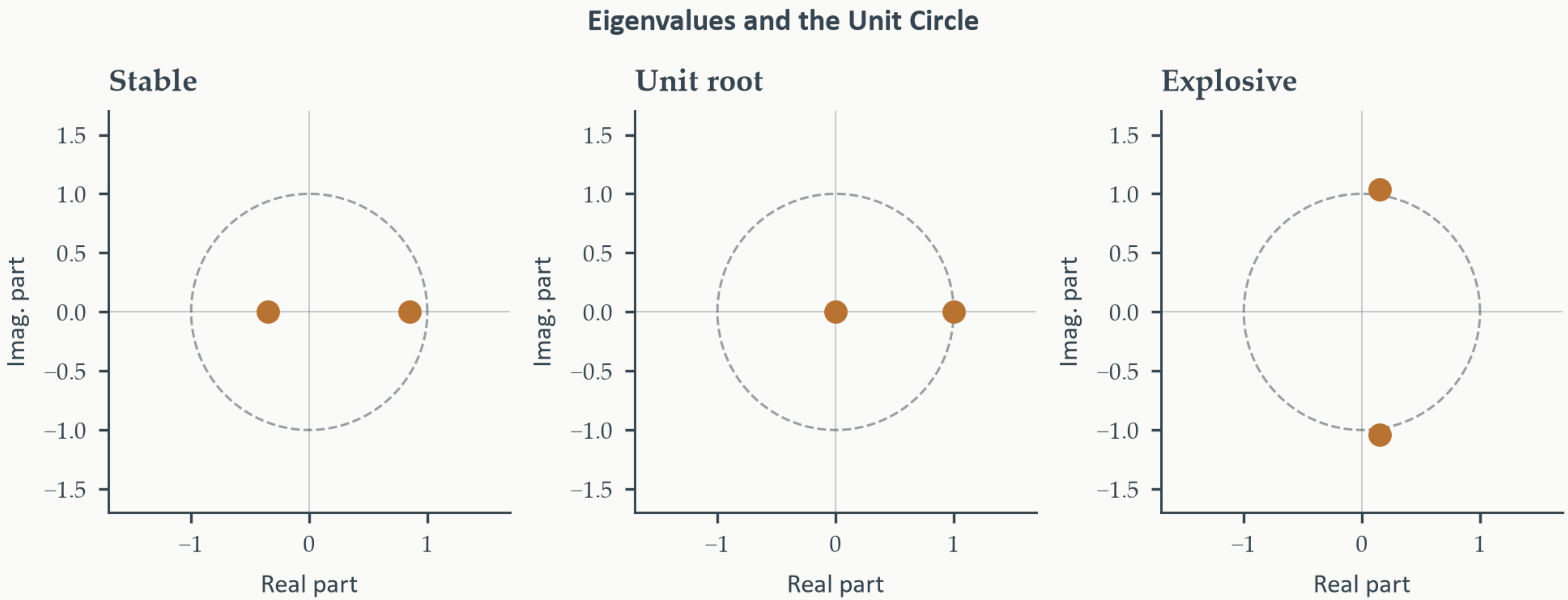
Difference equations

- Imaginary roots
  - $\lambda = \frac{\phi_1 \pm \sqrt{\phi_1^2 - 4\phi_2}}{2} = (\lambda_1, \lambda_2)$
  - What happens if $\phi_1^2 - 4\phi_2 < 0$?
  - You have imaginary roots, and they come in pairs
    - $\lambda_{1,2} = re^{\pm i\omega}$
    - r: modulus
    - ω: $P = \frac{2\pi}{\omega}$ periods per full oscillation

Difference equations

- In general terms
  - A process with p lags has p roots
  - Stability condition
    - All roots must fall inside the unit circle
    - $|\lambda_1|, \dots, |\lambda_p| < 1$
  - If **at least** one root = 1 → unit root process
  - If **at least** one root > |1| → explosive behavior
  - If there is an imaginary component, this tracks the cycle period

Difference equations



The ACF and PACF

Section 4

The ACF and the PACF

- ACF: Autocorrelation function
  - The persistence of the memory of $\{y_t\}$
  - Start with the variance of order $\gamma(h)$
    - $\gamma(0) = Cov(y_t, y_t) = \mathbb{E}[(y_t - \bar{y})(y_t - \bar{y})] = Var(y_t) = \sigma^2$
    - $\gamma(1) = Cov(y_t, y_{t-1}) = \mathbb{E}[(y_t - \bar{y})(y_{t-1} - \bar{y})]$
    - $\gamma(2) = Cov(y_t, y_{t-2}) = \mathbb{E}[(y_t - \bar{y})(y_{t-2} - \bar{y})]$

    - $\gamma(h) = Cov(y_t, y_{t-h}) = \mathbb{E}[(y_t - \bar{y})(y_{t-h} - \bar{y})]$
  - Then normalize by $\gamma(0)$
    - $\rho(h) = \frac{\gamma(h)}{\gamma(0)}$

The ACF and the PACF

- ACF: Autocorrelation function (cont…)
  - If we expect memory loss with a larger time "distance"
  - $|\rho(o)| > |\rho(1)| > |\rho(2)| > \cdots > |\rho(h)|$

  - The larger ϕ_i are, the slower the "decay" in the ACF values.

The ACF and the PACF

- PACF: Partial autocorrelation function
  - ACF: $y_{t-2} \rightarrow y_{\mathrm{t}-1} \rightarrow \mathrm{y_t}$
  - PACF: $y_{t-2} \rightarrow y_t$

  - The ACF tracks the memory effect on time. How much of y_{t-2} affects y_t depends on (a) an indirect effect **through** y_{t-1} and (b) a direct effect directly to y_t
  - The PACF tracks direct effects **only**. The direct impact of y_{t-2} on y_t net of what is channeled through y_{t-1}

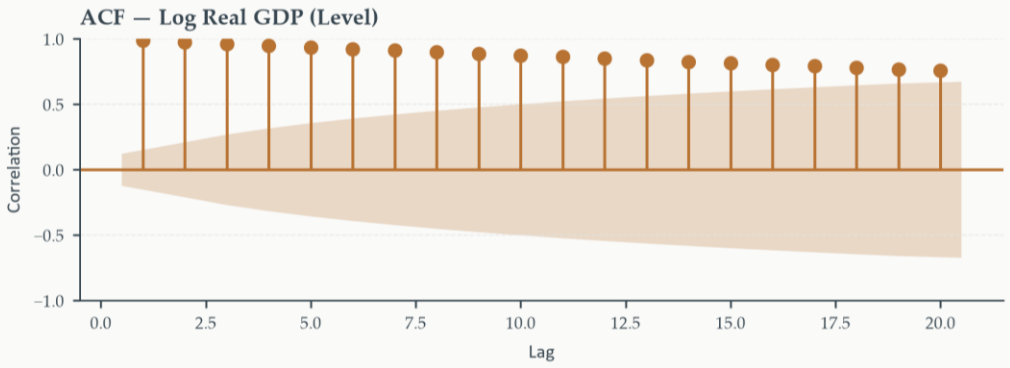
The ACF and the PACF

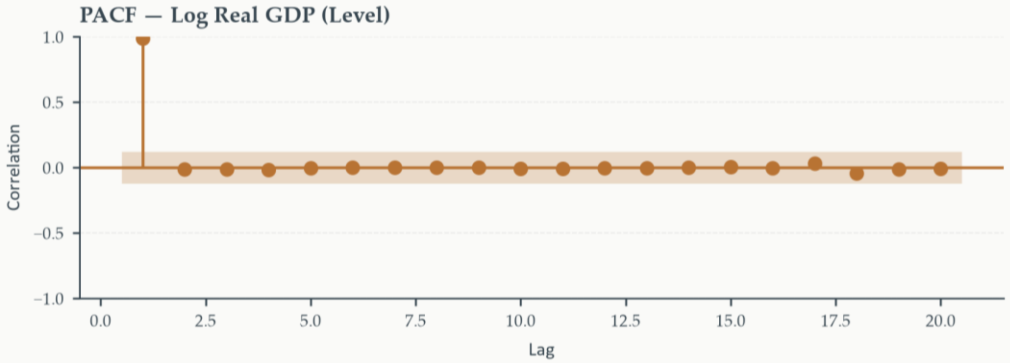
- PACF: Partial autocorrelation function (cont…)
  - Let $\alpha(h)$ be the PACF of lag h: $y_{t-h} \rightarrow y_t$
  - In lag = there is nothing to partial out, then ACF = PACF
    - $\alpha(1) = \rho(1)$
  - Lag 2 (and beyond) needs to be "cleaned" of any indirect effects
    - $\alpha(2) = \frac{\rho(2)-\rho(1)^2}{1-\rho(1)^2}$
  - Where do the $\alpha(h)$ come from?
    - The Yule-Walker equations (covered in chapter 3).

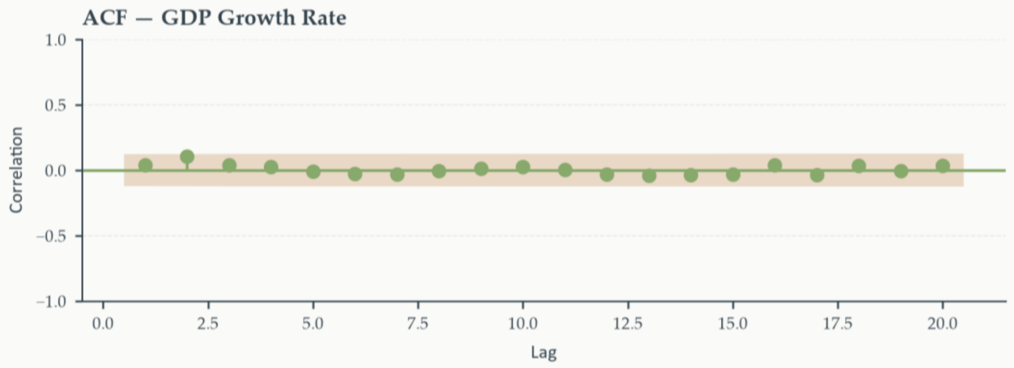
The ACF and the PACF

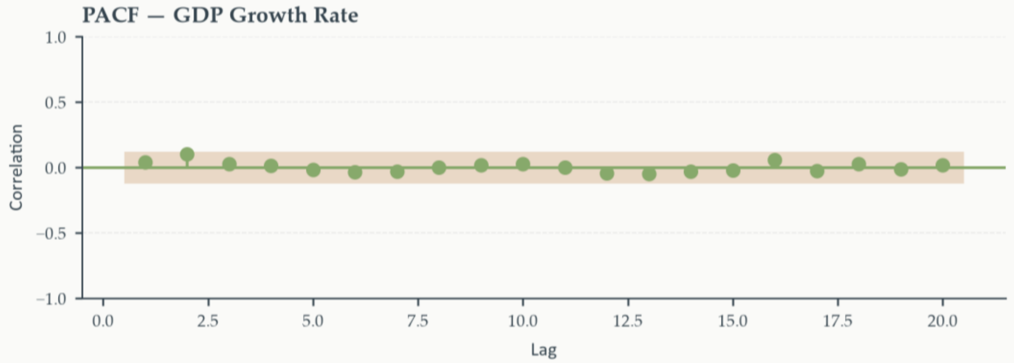
- ACF and PACF pattern recognition
  - Unit root: Slowly decaying ACF; close to 1 for many periods
  - Geometrically decaying: Autoregressive, AR(p), process
  - Sharp cutoff after a few lags: Moving average, MA(q), process
  - All bars inside the confidence bands: White Noise

The ACF and the PACF









Stationarity

Section 5

Stationarity

- **Stationarity** is the condition that a process has a stable probabilistic structure over time. Not that it is constant.
- Strict stationarity
  - A stochastic process $\{y_t\}$ is **strictly stationary** if any joint distribution $\left(y_{t_1}, \dots, y_{t_k}\right)$ has the same join distribution of $(y_{t_1+h}, \dots, y_{t_k+h})$ for any h
  - In simple terms, all observations come from the same probability distribution

Stationarity

- Weak (covariance) stationarity
  - If data has similar probability properties
  - A stochastic process $\{y_t\}$ is **weakly stationary** if:
    - (1) $\mathbb{E}[y_t] = \mu < \infty \forall t$ — the mean exists and is finite
    - (2) $Var(y_t) = \sigma^2 \forall t < \infty$ — the variance is finite and constant over time
    - (3) $Cov(y_t, y_{t-h}) = \gamma(h) \forall t, h$ — The covariance between any two observations depends only on the lag h, not on when in time they occur.
  - Condition (1): Rules out trending means
  - Condition (2): Rules out changing variance
  - Condition (3): Memory is stable and time-independent

Stationarity

- The unit root process
  - Unit root: When $\phi = -1$ or $\phi = 1$
  - When $\phi = 1$, $\bar{y} = \frac{c}{1-\phi}$ does not exist
  - Try $y_t^{(h)} = kt$
    - $y_t = \phi y_{t-1} + c$
    - $kt = k(t-1) + c$
    - $k = c$
  - Then
    - $y_t = \phi y_{t-1} + c$
    - $y_t = y_0 + ct$ (can you prove this?)

Stationarity

- The unit root process
  - Unit root: When $\phi = -1$ or $\phi = 1$
  - When $\phi = 1$, $\bar{y} = \frac{c}{1-\phi}$ does not exist
  - Try $y_t^{(h)} = kt$
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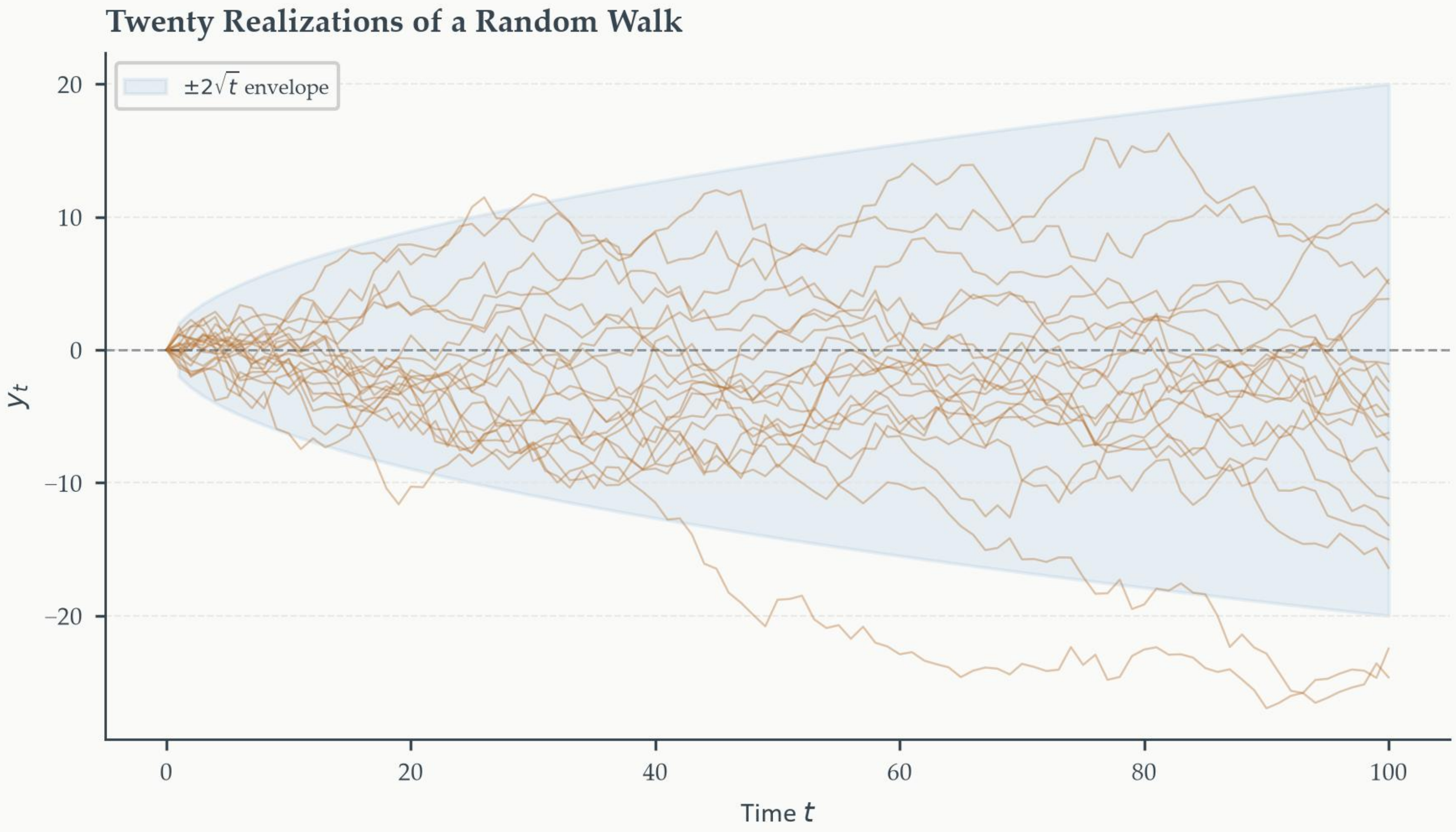
    - $kt = k(t-1) + c$
    - $k = c$
  - Then
    - $y_t = \phi y_{t-1} + c$
    - $y_t = y_0 + ct$ (can you prove this?)

Stationarity

- The unit root process
  - Two versions
    - $y_t = y_0 + ct$
    - If $c = 0$: pure RW
    - If $c \neq 0$: RW with drift linear trend
  - Add WN and take the first difference
    - $y_t = y_{t-1} + c + \varepsilon_t$
    - $\Delta y_t = c + \varepsilon_t$ (if $c \neq 0$)
    - $\Delta y_t = \varepsilon_t$ (if $c = 0$)

Stationarity



Testing for and fixing stationarity

Section 6

Testing for and fixing stationarity

- Start with a visual inspection of your data
- Trend-stationary and difference-stationary may look identical in a “small” sample
- When in doubt, run a formal test
  - ADF
  - KPSS

Testing for and fixing stationarity

- The ADF Test
  - $y_t = \phi y_{t-1} + c + \varepsilon_t$
  - Take the difference
  - DF Test
    - $\Delta y_t = \delta y_{t-1} + c + \varepsilon_t, \quad \delta = \phi - 1$
    - Now test on $H_0: \delta = 0$

Testing for and fixing stationarity

- The ADF Test (cont…)
  - The DF test may need to be augmented:
    - To account for a trend
    - To correct for serial correlation
  - The ADF test
    - $\Delta y_t = \delta y_{t-1} + \sum_{j=1}^{p} \psi_j \Delta y_{t-j} + c + \beta t + \varepsilon_t$
    - $\tilde{\varepsilon}_t = \varepsilon_t + \sum_{j=1}^{q} \theta_{t-t} \varepsilon_{t-j}\,, \quad \varepsilon_t \sim WN(0,1)$

Testing for and fixing stationarity

- The ADF Test (cont…)
  - Which ADF specification to use?
  - Let the plot tell you how to specify the test

Specification	Terms	Use when…
No constant, no trend	None	No drift and zero mean
Constant only	c	No drift, non-zero mean
Constant + trend	$(c + \beta t)$	Trend is present

Testing for and fixing stationarity

- The ADF Test (cont…)
  - How you pick the number p of lags? $\left(\sum_{j=1}^{p} \psi_j \Delta y_{t-j}\right)$
    - Start with a “large” p
    - Is that p-lagged coefficient statistical significant? No
    - Run the test with $p - 1$ lags
    - Repeat until the last lagged coefficient becomes statistically significant.

Testing for and fixing stationarity

- The KPSS Test
  - Reverse the null hypothesis
  - $y_t = \beta t + c_t + \varepsilon_t, \quad r_t = a r_{t-1} + \epsilon_t$
  - Then, test $H_0: Var(\epsilon_t) = 0$
    - Why?
    - If $Var(\sigma_t) = 0 \rightarrow c_t$ is constant and y_t is trend-stationary
  - The test statistic
    - $KPSS = \frac{1}{T^2} \frac{\sum_{t=1}^{T} S_t^2}{\hat{\lambda}^2}$
    - $S_t = \sum_{s=1}^{t} e_s$
    - $\hat{\lambda}^2$ is a long-run variance estimator that corrects for serial correlation

Testing for and fixing stationarity

- Because the ADF and KPSS tests have inverse null hypotheses, it is convenient to run both tests

ADF	KPSS	Joint conclusion
Reject H_0	Fail to reject H_0	**Both agree:** stationarity
Fail to reject H_0	Reject H_0	**Both agree:** unit root
Reject H_0	Reject H_0	Contradictory: Near integrated or structural break
Fail to reject H_0	Fail to reject H_0	Contradictory: Insufficient information

Testing and fixing stationarity

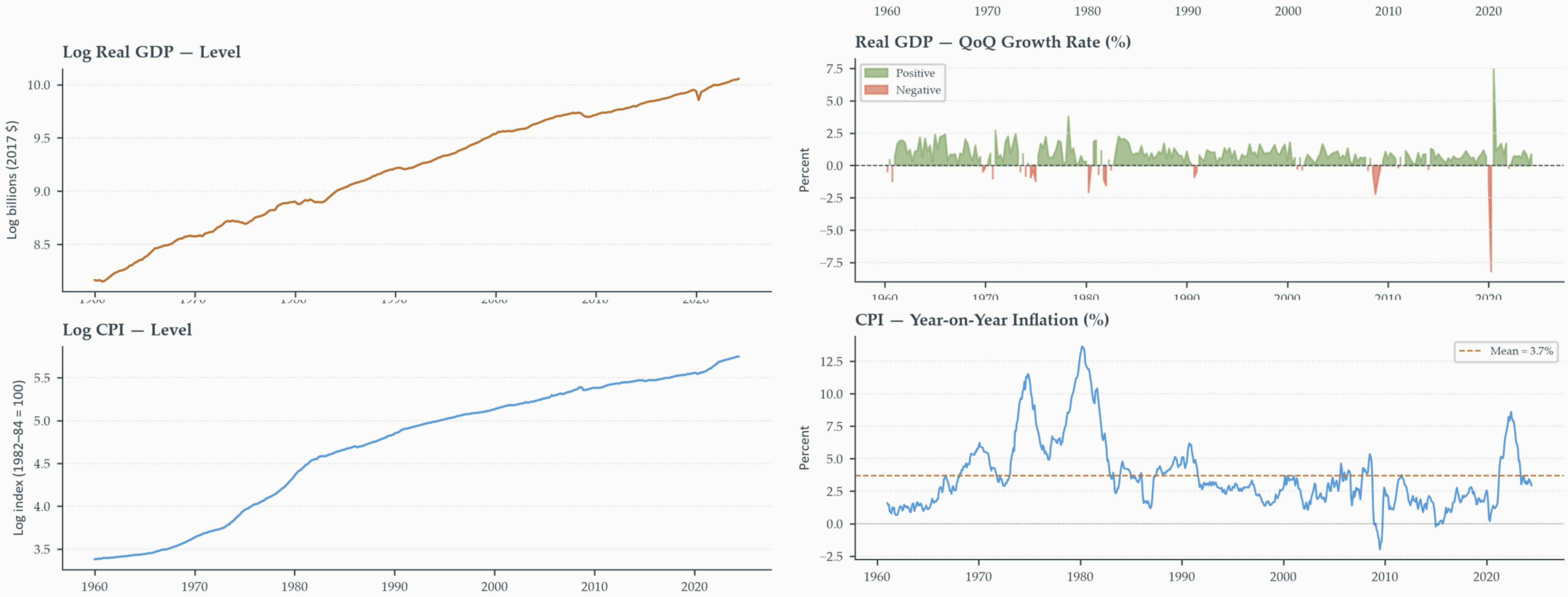
Trend stationary

- $y_t = \alpha + \beta t + \varepsilon_t$
  - Shocks are **transitory**
  - The series returns to its deterministic trend
  - Fix: Detrend

Difference stationary

- $y_t = \mu + y_{t-1} + \varepsilon_t$
  - Shocks are permanent
  - The series does not return to any deterministic path
  - Fix: Difference

Testing and fixing stationarity



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Chapter 1

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